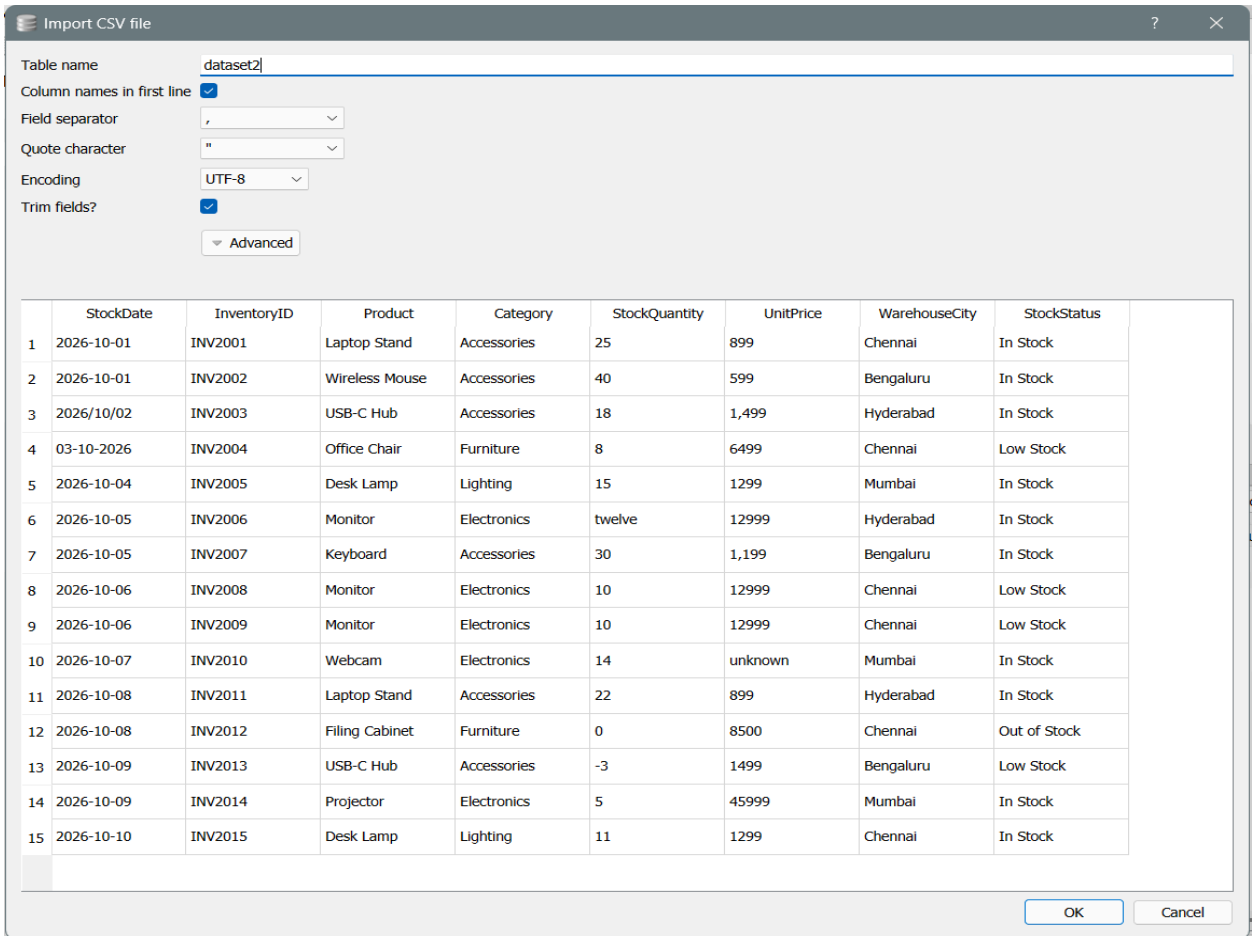
SQL DATA ANALYSIS REPORT

Dataset 2 — Electronics Orders

Dataset2.csv

```
StockDate,InventoryID,Product,Category,StockQuantity,UnitPrice,WarehouseCity,StockStatus
2026-10-01,INV2001,Laptop Stand,Accessories,25,899,Chennai,In Stock
2026-10-01,INV2002,Wireless Mouse,Accessories,40,599,Bengaluru,In Stock
2026/10/02,INV2003,USB-C Hub,Accessories,18,"1,499",Hyderabad,In Stock
03-10-2026,INV2004,Office Chair,Furniture,8,6499,Chennai,Low Stock
2026-10-04,INV2005,Desk Lamp ,Lighting,15,1299,Mumbai,In Stock
2026-10-05,INV2006,Monitor,Electronics,twelve,12999,Hyderabad,In Stock
2026-10-05,INV2007,Keyboard,Accessories,30,"1,199",Bengaluru,In Stock
2026-10-06,INV2008,Monitor,Electronics,10,12999,Chennai,Low Stock
2026-10-06,INV2009,Monitor,Electronics,10,12999,Chennai,Low Stock
2026-10-07,INV2010,Webcam,Electronics,14,unknown,Mumbai,In Stock
2026-10-08,INV2011,Laptop Stand,Accessories,22,899,Hyderabad,In Stock
2026-10-08,INV2012,Filing Cabinet,Furniture,0,8500,Chennai,Out of Stock
2026-10-09,INV2013,USB-C Hub,Accessories,-3,1499,Bengaluru,Low Stock
2026-10-09,INV2014,Projector,Electronics,5,45999,Mumbai,In Stock
2026-10-10,INV2015,Desk Lamp,Lighting,11,1299,Chennai,In Stock
```

Importing in SQLLite



	StockDate	InventoryID	Product	Category	StockQuantity	UnitPrice	WarehouseCity	StockStatus
1	2026-10-01	INV2001	Laptop Stand	Accessories	25	899	Chennai	In Stock
2	2026-10-01	INV2002	Wireless Mouse	Accessories	40	599	Bengaluru	In Stock
3	2026/10/02	INV2003	USB-C Hub	Accessories	18	1,499	Hyderabad	In Stock
4	03-10-2026	INV2004	Office Chair	Furniture	8	6499	Chennai	Low Stock
5	2026-10-04	INV2005	Desk Lamp	Lighting	15	1299	Mumbai	In Stock
6	2026-10-05	INV2006	Monitor	Electronics	twelve	12999	Hyderabad	In Stock
7	2026-10-05	INV2007	Keyboard	Accessories	30	1,199	Bengaluru	In Stock
8	2026-10-06	INV2008	Monitor	Electronics	10	12999	Chennai	Low Stock
9	2026-10-06	INV2009	Monitor	Electronics	10	12999	Chennai	Low Stock
10	2026-10-07	INV2010	Webcam	Electronics	14	unknown	Mumbai	In Stock
11	2026-10-08	INV2011	Laptop Stand	Accessories	22	899	Hyderabad	In Stock
12	2026-10-08	INV2012	Filing Cabinet	Furniture	0	8500	Chennai	Out of Stock
13	2026-10-09	INV2013	USB-C Hub	Accessories	-3	1499	Bengaluru	Low Stock
14	2026-10-09	INV2014	Projector	Electronics	5	45999	Mumbai	In Stock
15	2026-10-10	INV2015	Desk Lamp	Lighting	11	1299	Chennai	In Stock

Data Modifying and Cleaned Data Insertion:

```
DROP TABLE IF EXISTS retail_inventory;
CREATE TABLE retail_inventory (
 StockDate TEXT, InventoryID TEXT, Product TEXT, Category TEXT,
 StockQuantity TEXT, UnitPrice TEXT, WarehouseCity TEXT, StockStatus TEXT
);

INSERT INTO retail_inventory VALUES
('2026-10-01','INV2001','Laptop Stand','Accessories','25','899','Chennai','In Stock'),
('2026-10-01','INV2002','Wireless Mouse','Accessories','40','599','Bengaluru','In Stock'),
('2026/10/02','INV2003','USB-C Hub','Accessories','18','1,499','Hyderabad','In Stock'),
('03-10-2026','INV2004','Office Chair','Furniture','8','6499','Chennai','Low Stock'),
('2026-10-04','INV2005','Desk Lamp ','Lighting','15','1299','Mumbai','In Stock'),
('2026-10-05','INV2006','Monitor','Electronics','twelve','12999','Hyderabad','In Stock'),
('2026-10-05','INV2007','Keyboard','Accessories','30','1,199','Bengaluru','In Stock'),
('2026-10-06','INV2008','Monitor','Electronics','10','12999','Chennai','Low Stock'),
('2026-10-06','INV2009','Monitor','Electronics','10','12999','Chennai','Low Stock'),
('2026-10-07','INV2010','Webcam','Electronics','14','unknown','Mumbai','In Stock'),
('2026-10-08','INV2011','Laptop Stand','Accessories','22','899','Hyderabad','In Stock'),
('2026-10-08','INV2012','Filing Cabinet','Furniture','0','8500','Chennai','Out of Stock'),
('2026-10-09','INV2013','USB-C Hub','Accessories','-3','1499','Bengaluru','Low Stock'),
('2026-10-09','INV2014','Projector','Electronics','5','45999','Mumbai','In Stock'),
('2026-10-10','INV2015','Desk Lamp','Lighting','11','1299','Chennai','In Stock');
```

Query 1 — View all inventory records

```
SELECT rowid AS RowNumber, * FROM retail_inventory ORDER BY rowid;
```

Output:

	RowNumber	StockDate	InventoryID	Product	Category	StockQuantity	UnitPrice	WarehouseCity	StockStatus
1	1	2026-10-01	INV2001	Laptop Stand	Accessories	25	899	Chennai	In Stock
2	2	2026-10-01	INV2002	Wireless Mouse	Accessories	40	599	Bengaluru	In Stock
3	3	2026/10/02	INV2003	USB-C Hub	Accessories	18	1,499	Hyderabad	In Stock
4	4	03-10-2026	INV2004	Office Chair	Furniture	8	6499	Chennai	Low Stock
5	5	2026-10-04	INV2005	Desk Lamp	Lighting	15	1299	Mumbai	In Stock
6	6	2026-10-05	INV2006	Monitor	Electronics	twelve	12999	Hyderabad	In Stock
7	7	2026-10-05	INV2007	Keyboard	Accessories	30	1,199	Bengaluru	In Stock
8	8	2026-10-06	INV2008	Monitor	Electronics	10	12999	Chennai	Low Stock
9	9	2026-10-06	INV2009	Monitor	Electronics	10	12999	Chennai	Low Stock
10	10	2026-10-07	INV2010	Webcam	Electronics	14	unknown	Mumbai	In Stock
11	11	2026-10-08	INV2011	Laptop Stand	Accessories	22	899	Hyderabad	In Stock
12	12	2026-10-08	INV2012	Filing Cabinet	Furniture	0	8500	Chennai	Out of Stock
13	13	2026-10-09	INV2013	USB-C Hub	Accessories	-3	1499	Bengaluru	Low Stock
14	14	2026-10-09	INV2014	Projector	Electronics	5	45999	Mumbai	In Stock
15	15	2026-10-10	INV2015	Desk Lamp	Lighting	11	1299	Chennai	In Stock

Explanation: Displays the full inventory dataset for checking.

Q2. Data-quality review

```
SELECT rowid AS RowNumber, * FROM retail_inventory
WHERE StockDate NOT GLOB '????-??-??'
 OR TRIM(Product) <> Product
 OR StockQuantity = 'twelve'
 OR CAST(StockQuantity AS INTEGER) < 0
 OR UnitPrice = 'unknown'
 OR UnitPrice LIKE '%,%'
ORDER BY rowid;
```

Output:

	RowNumber	StockDate	InventoryID	Product	Category	StockQuantity	UnitPrice	WarehouseCity	StockStatus
1	3	2026/10/02	INV2003	USB-C Hub	Accessories	18	1,499	Hyderabad	In Stock
2	4	03-10-2026	INV2004	Office Chair	Furniture	8	6499	Chennai	Low Stock
3	5	2026-10-04	INV2005	Desk Lamp	Lighting	15	1299	Mumbai	In Stock
4	6	2026-10-05	INV2006	Monitor	Electronics	twelve	12999	Hyderabad	In Stock
5	7	2026-10-05	INV2007	Keyboard	Accessories	30	1,199	Bengaluru	In Stock
6	10	2026-10-07	INV2010	Webcam	Electronics	14	unknown	Mumbai	In Stock
7	13	2026-10-09	INV2013	USB-C Hub	Accessories	-3	1499	Bengaluru	Low Stock

Explanation: Finds inconsistent dates, extra spaces, nonnumeric or negative quantities, and suspicious prices.

Q3. Records by stock status

```
SELECT StockStatus, COUNT(*) AS ProductRecords
FROM retail_inventory GROUP BY StockStatus
ORDER BY ProductRecords DESC, StockStatus;
```

Output:

	StockStatus	ProductRecords
1	In Stock	10
2	Low Stock	4
3	Out of Stock	1

Explanation: Summarizes stock status counts.

Q4. Valid stock by category

```
SELECT Category, SUM(CAST(StockQuantity AS INTEGER)) AS TotalStock
FROM retail_inventory
WHERE StockQuantity GLOB '[0-9]*' AND CAST(StockQuantity AS INTEGER) > 0
GROUP BY Category ORDER BY TotalStock DESC;
```

Output:

	Category	TotalStock
1	Accessories	135
2	Electronics	39
3	Lighting	26
4	Furniture	8

Explanation: Adds only positive numeric quantities within each category.

Q5. Inventory value by city

```
SELECT WarehouseCity,
 SUM(CAST(StockQuantity AS INTEGER) * CAST(REPLACE(UnitPrice, ',', '') AS REAL)) AS InventoryValue
FROM retail_inventory
WHERE StockQuantity GLOB '[0-9]*' AND CAST(StockQuantity AS INTEGER) > 0
 AND UnitPrice NOT IN ('unknown', '')
GROUP BY WarehouseCity ORDER BY InventoryValue DESC;
```

Output:

	WarehouseCity	InventoryValue
1	Chennai	348736.0
2	Mumbai	249480.0
3	Bengaluru	59930.0
4	Hyderabad	46760.0

Explanation: Estimates stock value by city while excluding invalid quantities and unknown prices.

Q6. Repeated product names

```sql
SELECT TRIM(Product) AS Product, COUNT(*) AS RecordCount
FROM retail_inventory GROUP BY TRIM(Product)
HAVING COUNT(*) > 1 ORDER BY RecordCount DESC, Product;
```

Output:

	Product	RecordCount
1	Monitor	3
2	Desk Lamp	2
3	Laptop Stand	2
4	USB-C Hub	2

Explanation: Uses GROUP BY and HAVING to find product names appearing in multiple records.

Q7. Stock value bands

```sql
SELECT InventoryID, TRIM(Product) AS Product,
 CAST(StockQuantity AS INTEGER) * CAST(REPLACE(UnitPrice, ',', '') AS REAL) AS StockValue,
CASE
  WHEN CAST(StockQuantity AS INTEGER) * CAST(REPLACE(UnitPrice, ',', '') AS REAL) < 5000 THEN 'Low'
  WHEN CAST(StockQuantity AS INTEGER) * CAST(REPLACE(UnitPrice, ',', '') AS REAL) < 30000 THEN 'Medium'
  ELSE 'High'
 END AS ValueBand
FROM retail_inventory
WHERE StockQuantity GLOB '[0-9]*' AND CAST(StockQuantity AS INTEGER) > 0
 AND UnitPrice NOT IN ('unknown', '')
ORDER BY StockValue DESC;
```

Output:

	InventoryID	Product	StockValue	ValueBand
1	INV2014	Projector	229995.0	High
2	INV2008	Monitor	129990.0	High
3	INV2009	Monitor	129990.0	High
4	INV2004	Office Chair	51992.0	High
5	INV2007	Keyboard	35970.0	High
6	INV2003	USB-C Hub	26982.0	Medium
7	INV2002	Wireless Mouse	23960.0	Medium
8	INV2001	Laptop Stand	22475.0	Medium
9	INV2011	Laptop Stand	19778.0	Medium
10	INV2005	Desk Lamp	19485.0	Medium
11	INV2015	Desk Lamp	14289.0	Medium

Explanation: Uses CASE to classify estimated inventory value as Low (<5,000), Medium (5,000–29,999.99), or High (30,000+).

Q8. Inventory records by city and category

```sql
SELECT WarehouseCity, Category, COUNT(*) AS RecordCount
FROM retail_inventory GROUP BY WarehouseCity, Category
ORDER BY WarehouseCity, RecordCount DESC, Category;
```

Output:

	WarehouseCity	Category	RecordCount
1	Bengaluru	Accessories	3
2	Chennai	Electronics	2
3	Chennai	Furniture	2
4	Chennai	Accessories	1
5	Chennai	Lighting	1
6	Hyderabad	Accessories	2
7	Hyderabad	Electronics	1
8	Mumbai	Electronics	2
9	Mumbai	Lighting	1

Explanation: Compares how inventory records are distributed across cities and categories.